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Paleogeography of India at ~1.8 Ga: New constraints from baddeleyite geochronology and paleomagnetism of mafic dykes from the Dharwar craton

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A B S T R A C T

The paleogeographic position of Dharwar craton/India within Paleoproterozoic supercontinent configurations has been well demonstrated using highly reliable key paleopoles from ~2.37 Ga to ~1.86 Ga mafic events in recent studies. Here, we report a paleomagnetic pole at ~1.8 Ga along with a precise Pb-Pb baddeleyite age. The emplacement age of an NW-SE trending dolerite dyke yields a weighted mean age of 1799 ± 0.8 Ma (MSWD = 0.7), which coincides with earlier reported U-Pb baddeleyite age (1794 ± 7 Ma) and confirms the existence of significant emplacement of Pebbair dyke swarm at ~1.8 Ga. Paleomagnetic results from eleven sites shows mean D = 306°, I = 31°, α_{95} = 10° (k = 23), and corresponding paleopole is calculated at 38°N, 357°E (dp = 6.1° and dm = 11°). These remanent directions are carried by the pseudo-single domain (PSD) magnetite. This pole passes the paleosecular variation test of the geomagnetic field and fulfills the reliability criteria (R = 5) of Meert et al. (2020), so, suggested to be a key pole for Dharwar craton/ India (~1.8 Ga). The paleopositon of Dharwar craton is occupied at shallow paleolatitude and drifted towards the north with a minimum velocity (~5.7 cm/yr) from ~1.86 Ga to ~1.80 Ga. The paleogeographic reconstruction model at ~1.8 Ga has been attempted with available coeval paleopoles from other global cratons. The Dharwar craton (South India block), the Amazonia craton, and the Baltica craton have assembled at ~1.86 Ga, and North Australia, Sao Francisco, Rio de la Plata cratons came in close to the South India block, Amazonia, and Baltica assembly at ~1.80 Ga.

1. Introduction

Mafic dykes are normally regarded as exceptional regional time markers and magnetic recorders due to their rapid emplacement on the subsurface or surface levels, derived from mantle sources over great crustal depths (Halls, 1982). Hence, the determination of precise age, coupled with a primary magnetization of these dykes, can provide vital information for understanding the timing of plume activity, paleogeographic position in ancient supercontinent configurations, and associated tectonics (Bleeker and Ernst, 2006; Ernst et al., 2013; Ernst and Jowitt 2013; Pisarevsky et al., 2013, 2014; Kumar et al., 2017; Shellnutt et al., 2018; Shankar et al., 2018; Babu et al., 2018; Parashuramulu et al., 2021; Pesonen et al., 2021; Evans et al., 2021; Elming et al., 2021; Meert and Santosh, 2022; Shankar et al., 2023; Panda et al., 2023). Several extensive mafic dyke swarms were identified in many Precambrian terrains, commonly serving as plumbing systems for Large Igneous Province (LIP) flood basalts (Ernst and Bleeker 2010; Kumar et al., 2017; Buchan and Ernst 2018; Ernst et al., 2019).

The Dharwar Craton (DC) of South India is a large Archean craton, that has experienced extensive crustal expansion and emplacement of mafic dykes during the Paleoproterozoic (Drury, 1984; Murthy et al., 1987; Halls et al., 2007; French et al., 2008; French and Heaman 2010; Radhakrishna et al., 2013; Belica et al., 2014; Söderlund et al., 2019; Sarma et al., 2020; Parashuramulu et al., 2021; Miller et al., 2023). These mafic dykes are widely dispersed throughout the craton with distinct trends and dimensions (French and Heaman 2010; Radhakrishna et al., 2013; Belica et al., 2014). Eight discrete major mafic dyke swarms have been identified during Paleoproterozoic with robust age constraints including the Karimnagar-Bangalore giant dyke swarm (~2369–2361 Ma; Halls et al., 2007; French and Heaman 2010; Kumar et al., 2012a; Söderlund et al., 2019; Babu et al., 2018, 2020; Sarma et al., 2020), the Ippaguda dyke swarm (~2257–2253 Ma; Nagaraju et al., 2018b; Söderlund et al., 2019), the AKLD swarm (~2221–2212 Ma; French and Heaman 2010; Srivastava et al., 2011; Kumar et al., 2012b, 2014;; Nagaraju et al., 2018a; Söderlund et al., 2019; Nagaraju and Parashuramulu, 2019), the Narapally-Arehalli dyke swarm (~2213–2205 Ma; Nagaraju et al., 2018b; Yadav et al., 2020; Yadav and Sarma 2021), the Cuddapah dyke swarm (~2087–2081 Ma; Kumar et al., 2015; Parashuramulu, 2017; Söderlund et al., 2019; Parashuramulu et al., 2023), the Mahabubnagar- Dandeli dyke swarm (~2181–2177 Ma; French and Heaman, 2010), the Hampi dyke swarm (~1894–1881 Ma; Halls et al., 2007; French et al., 2008; Sarma et al.,

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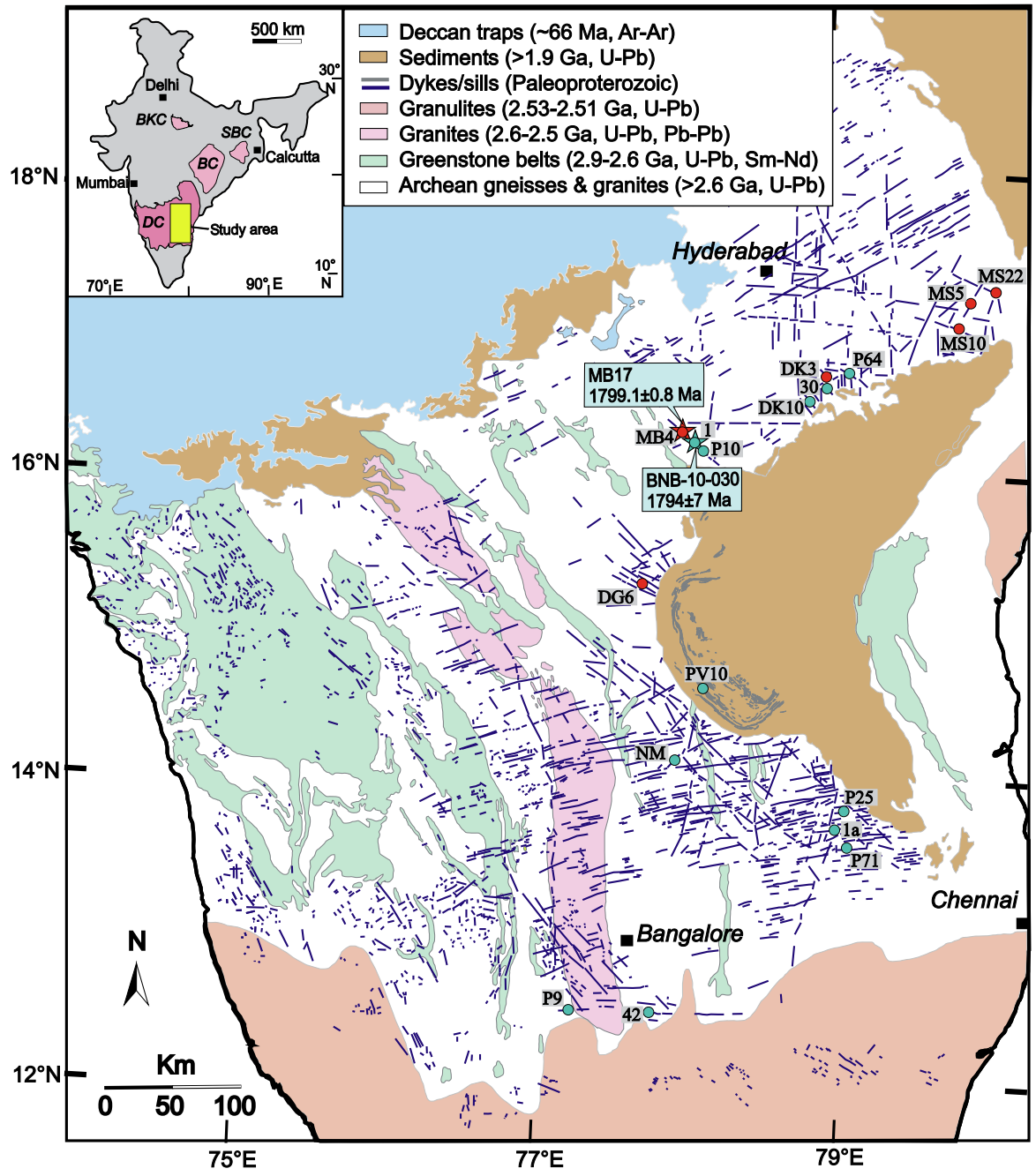


Fig. 1. Geological map of the Dharwar craton showing the distribution of mafic dykes and sampling site locations. Paleomagnetic sites in the present study shown with red circles and blue circles represent reported paleomagnetic sites. Pb-Pb baddeleyite geochronology (present study) site represented by a red star and blue star displaying U-Pb baddeleyite age (published).

2020); the SangalaPalli- Chimiryala swarm (~1867–1862 Ma; Sarma et al., 2020; Parashuramulu et al., 2021).

In addition, a U-Pb baddeleyite age of 1794 ± 7 Ma for a NW-SE trending dyke from the northwestern region of the Cuddapah basin has been reported (Söderlund et al., 2019). This mafic event represents a distinctly youngest generation of mafic magmatic activity in the Paleoproterozoic history (based on available precise U-Pb/Pb-Pb ages) of the DC. A comparable slightly younger Pb-Pb baddeleyite age of 1765.3 ± 1.0 Ma (weighted mean age) for the Pipilia dyke swarm (Shankar et al., 2014, 2018) was recognized in the Singhbhum Craton. However, the Paleoposition of India at about ~1.8 Ga and its relationship with neighboring continents in supercontinent reconstructions are uncertain due to the scarcity of primary paleomagnetic poles. This study is initiated to obtain a precise paleopole for the ~1.8 Ga Pebbair dyke swarm by an integrated study of TE-TIMS Pb-Pb baddeleyite age and paleomagnetic data from dykes within the Dharwar craton. Based on our new results and published coeval key paleomagnetic poles from other continents a possible paleogeographic reconstruction at ~1.8 Ga is presented in an attempt to define relative positions at that time.

2. Geologic setting and sampling

There are several well-preserved Archean cratons in the Indian subcontinent, i.e. the Aravalli, the Bastar, the Bundelkhand, the Dharwar, and the Singhbum (Fig. 1). Most of the southern peninsula of India is part of the Dharwar craton. The Dharwar craton is overlain by the

Table 1
TE-TIMS Pb isotopic data on baddeleyite fractions from the Pebbair dyke, Dharwar craton.

Sample. No.	No. bloc.	$^{206}Pb/^{204}Pb$ (m)	Abs. err	$^{207}Pb/^{206}Pb$ (m)	$^{207}Pb/^{206}Pb$ (c)	Age (Ma)
MB17 (16.258°N; 77.940°E), Strike: N121°E						
MB17-1	25	8843	661	0.111782 ± 020	0.109926 ± 079	1798.2 ± 2.3
MB17-2	25	5405	98	0.112434 ± 092	0.109898 ± 084	1797.7 ± 2.4
MB17-3	30	5225	101	0.112590 ± 068	0.109969 ± 057	1798.9 ± 1.9
MB17-4	21	4094	25	0.113300 ± 068	0.109979 ± 064	1799.1 ± 2.0
MB17-5	27	5460	48	0.112515 ± 050	0.110025 ± 048	1799.8 ± 1.8
MB17-6	27	5004	101	0.112766 ± 066	0.110026 ± 049	1799.8 ± 1.8
Weighted Mean age = 1799.1 ± 0.8 Ma, MSWD = 0.68						

$^{206}Pb/^{204}Pb$ (m) and $^{207}Pb/^{206}Pb$ (m) are measured values (fraction means with standard error) and $^{207}Pb/^{206}Pb$ (c) is the corrected value. Uncertainties are $2\sigma_m$ and refer to the least significant digits. Age errors include fractionation uncertainty of 0.055%. Weighted mean ages were calculated using Ludwig (2003).

Table 2
Paleomagnetic results on ~1.8 Ga Pebbair dyke swarm from the Dharwar craton, India.

Site	Lat (°N)	Long (°E)	Strike (°E)	n/N	D (°)	I (°)	k	α_{95}	Plat (°N)	Plong (°E)	dp	dm	Ref.
MB4*	16.2581	77.9398	121	10	306.8	32.0	16	12.3	39.2	357.3	7.8	13.9	1
MS5	17.1344	79.9151	45	5	301.9	42.7	201	6.5	35.6	8.5	5.0	8.0	1
MS22	17.2361	80.0928	–	10	293.0	39.4	32	8.7	27.2	6.8	6.2	10.4	1
MS10	16.9702	79.8718	118	10	289.1	27.9	34	8.4	22.1	358.4	5.0	9.1	1
DK3	16.6778	78.9148	129	10	298.2	35.4	34	9.7	31.5	2.0	6.4	11.2	1
DG6	15.2942	77.6866	127	6	325.5	43.9	29	12.7	56.2	11.3	9.9	15.9	1
1*	16.2161	77.9960	100	22	298.0	17.0	64	5.8	29.3	348.7	3.0	6.0	2
30	16.5473	78.8905	110	11	306.0	15.0	16	11.7	36.1	346.2	6.2	12.0	2
DK10	16.537	78.808	335	7	293.0	18.0	93	6	24	352	3.0	7.0	3
PV10#	14.607	78.041	–	8	293.0	19.0	71	7	24	352	4	7	3
NM#	14.138	77.935	60	5	314.5	18.8	15	20.3	45.3	347.5	21.1	11	4
P10*#	16.230	78.010	NW	6	335.7	21.0	20	15.4	65.8	338.3	16.2	8.5	5
P64#	16.980	79.070	NW	4	308.5	22.8	14	25.6	40.0	350.7	27.2	14.4	5
P25#	13.790	79.070	E-W	8	325.6	17.8	14	15.5	56.0	344.9	16.1	8.3	5
P71#	13.580	79.090	NE	5	333.2	17.9	19	17.9	63.4	342.4	18.6	9.6	5
42	12.4647	77.7673	160	8	336	21	170	3	68.0	346	3.2	1.7	6
1a	13.6526	78.9659	80	5	320	41	16	13	53.0	11	15.8	9.6	6
P9#	12.424	77.234	E-W	6	304.6	17.3	13	16.1	35.5	349.3	16.7	8.6	7
Grand Mean	**15.500**	**77.500**	——	**104/11**	**305.8**	**31.1**	**23**	**9.7**	**38.0**	**356.9**	**6.1**	**10.9**	

Lat (°N) and Long (°E) are site latitude and site longitude respectively; n/N is the number of samples from each site and number of sites contributing to the mean; S(°) is strike of the dyke at the respective sites; D(°) and I(°) are the mean declination and mean inclination; k is the Fisher's (1953) precision parameter of the mean pole; α_{95} is the radius of the circle of 95% confidence of the mean directions; Plat and Plong are the palaeolatitude and palaeolongitude for the virtual geomagnetic pole; dp and dm are the semi-axes of the oval of 95% confidence. Grand mean is calculated using central location in the Dharwar craton (15.5°, 77.5°); *: site with precise age; #: Site which is not considered for mean calculation; 1: Present study; 2: Miller et al., 2023; 3: Parashuramulu et al., 2021; 4: Belica et al., 2014; 5: Radhakrishna et al., 2013; 6: Pivarunas et al., 2018; 7: Piispa et al., 2011.

Cretaceous Deccan basalts in the north and northwest and is truncated by the Godavari rift to the northeast. In the east, the craton is bounded by the ~1.5 Ga Eastern Ghats mobile belt and on the west by the Arabian Sea.

The Archean crust as summarized by Swami Nath and Ramakrishnan (1981), consists of a southern granulite block and a low-grade metamorphic northern block. The northern block has been further divided into a low-temperature metamorphic western (Western Dharwar Craton, WDC) and a high-temperature eastern block (Eastern Dharwar Craton, EDC), separated by an N-NW trending fault zone. The amalgamation of the western and eastern Dharwar cratonic blocks is interpreted to have taken place at ~2.5 Ga (Chadwick et al., 2000; Drury, 1983; Balakrishnan et al., 1999). The older WDC (3.4–2.7 Ga) predominantly consists of tonalite-trondhjemite-granodiorite (TTG) gneissic basement rocks overlaid by green stone belts (Chadwick et al., 2000, Jayananda et al., 2006, 2013, 2018; Mohan et al., 2014, 2023; Sarma et al., 2012). The younger EDC (3.0–2.5 Ga) is composed of late Archaean (2.6–2.5 Ga) granites intruded into subordinate amounts of older (2.9–2.7 Ga) TTG gneisses (Chadwick et al., 2000; Mohan et al., 2013; Saikrishna et al., 2023). Neoarchean greenstones in the EDC are confined to narrow and elongated belts (more or less North–South trending), which may represent terrane boundaries (Krogstad et al., 1989; Chadwick et al., 2000).

A total of 51 oriented block samples (at 6 sites) were collected from six mafic dykes in the northern and northwestern fringes of the Cuddapah basin following standard protocols of sampling. Site geographical coordinates are given in Table 2 and shown in Fig. 1. The samples were oriented using sun and magnetic compasses. All the sampled dykes are dolerite in composition and show vertical or subvertical dips. These dykes transect Archean basement rocks and are exposed as fresh outcrops in quarries, road, and canal cuttings. At least five block samples were collected at each site and then drilled into cylindrical cores followed by cutting into standard-size specimens. In addition, one coarse-grained sample (MB17) was also collected from an NW-SE trending dyke (site MB4) for Pb-Pb baddeleyite dating. A U-Pb baddeleyite age (1794 ± 7 Ma; Söderlund et al., 2019) has also been reported on the extension of this dyke near Pebbair town.

3. Methodology

3.1. Baddeleyite geochronology

One coarse-grained sample (MB17) was selected for baddeleyite mineral separation and Pb-Pb dating. The sample (~1 kg) was crushed to fine rock powder (<125-μm grain size) using a jaw crusher followed by a ring grinder, and rock powder was loaded on a ''water-based'' mineral separation technique using a Wilfley table followed by Söderlund and Johansson (2002) procedure. High-quality (free of mineral inclusions, fractures, and alteration) baddeleyite crystals were carefully handpicked (at least 100 grains) under a binocular microscope

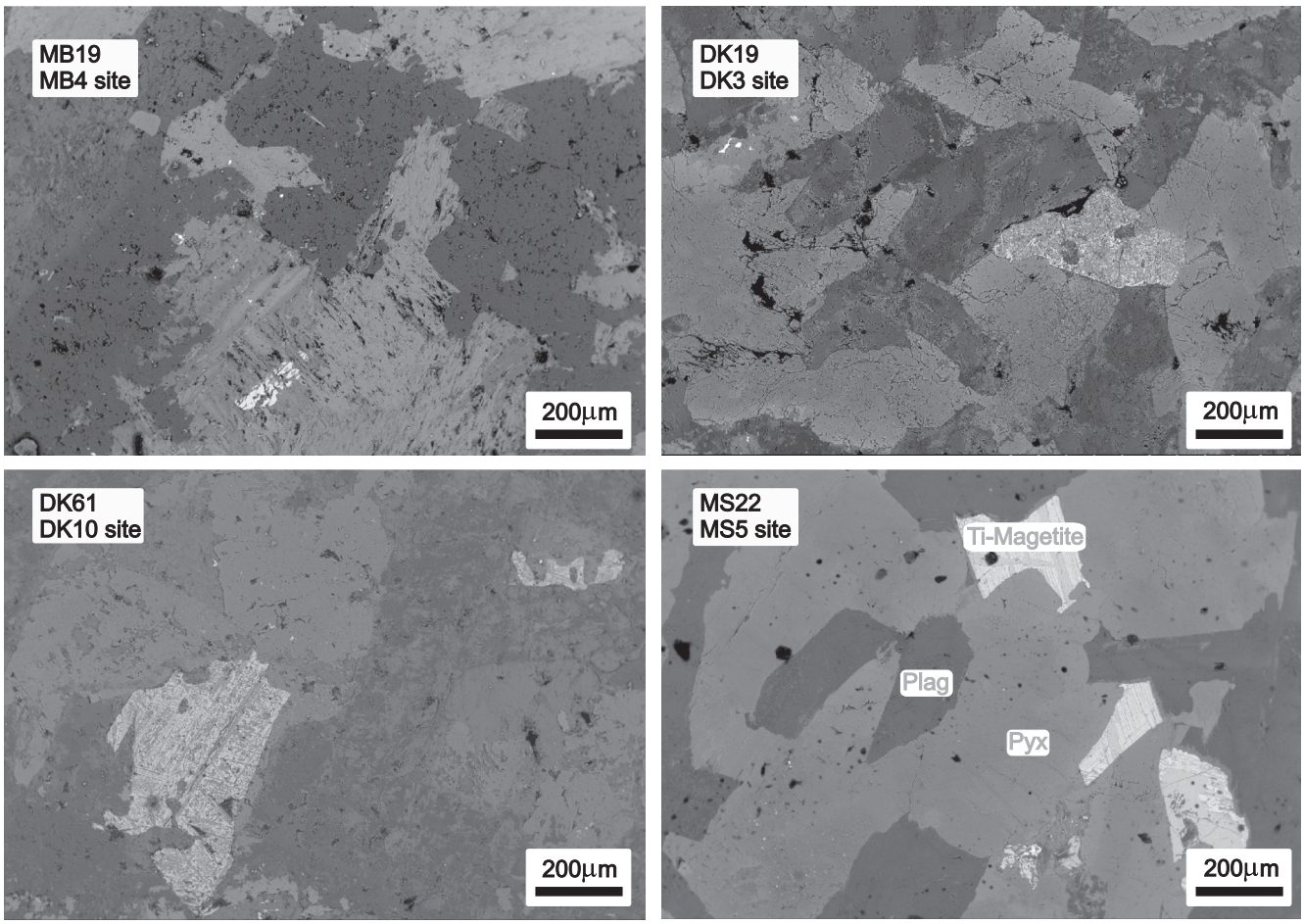


Fig. 2. Scanning electron microscopic photographs are showing a mineral assemblage of representative samples from ~1.8 Ga Pebbair dyke swarm. Plag: Plagioclase feldspars; Pyx: Pyroxenes.

from collected heavy mineral fraction. These baddeleyite grains were cleaned in 4N HNO_3 at room temperature, and grains were rinsed several times in MQ H_2O and dried. These cleaned baddeleyite grains were loaded on degassed rhenium (Re) filaments, which were bent in a U' shaped valley in the center. The grains were placed at the bottom of the valley and kept in place with a minute droplet of silica gel for annealing at 1250 °C in a vacuum chamber at ~$1 * 10^{-7}$ mbar for 30 min to expel disturbed Pb from altered domains. Grains were subsequently embedded in silica glass on the same Re filament by fusing the sample with silica gel and phosphoric acid mixture at 1200 °C for ~10 min. in a vacuum chamber. The silica gel phosphoric acid mixture used was 2:1 of concentrated silicic acid and 9 N phosphoric acid. The maximum amount of silica gel mixture used was restricted to <1 Âµl, to keep blank to a minimum (<2 picograms). Sample loading was performed under a stereomicroscope in a clean air station fitted with a HEPA filter. Analyses were performed on a TRITON^{PLUS} Thermal Ionization Mass Spectrometer at the CSIR-National Geophysical Research Institute, Hyderabad. The three high-mass Faraday detectors (H1 to H3) connected to 10^{11}-ohm resistors were used to measure ^{206}Pb, ^{207}Pb, and ^{208}Pb, in static multi-collector mode with virtual amplifier rotation. Details regarding data acquisition and data reduction are given in Kumar et al. (2014).

3.2. Paleomagnetism

Paleomagnetic measurements were carried out in the Paleomagnetism division at CSIR-National Geophysical Research Institute, Hyderabad, India. To isolate the characteristic remanent magnetization (ChRM) directions step-wise alternating magnetic field (AF) and thermal (TH) demagnetizing techniques were applied. Stepwise AF-demagnetization using intervals of 2.5 mT (between 0 and 20 mT) followed by 10 mT intervals thereafter. Thermally demagnetized samples used discrete 50 °C steps up to 500 °C followed by 20 °C steps thereafter. An AGICO JR-6 spinner magnetometer with a sensitivity of $2.4*10^{-6}$ A/m was used to isolate the components of magnetization for each sample. Magnetic components were visually analyzed using Zijderveld plots (Zijderveld, 1967), and calculated by principal component analysis (PCA) (Kirschvink, 1980). A minimum of five demagnetization steps were used to define vectors by least-squares fits, and a maximum of ≤10° was used for mean angular deviation (MAD). Site mean directions and paleomagnetic poles were calculated using Fisher's (1953) statistics. In addition, we have conducted rock magnetic experiments on one rock-powered sample from each dyke to identify the carriers of magnetic remanence in the samples. Thermomagnetic curves (susceptibility vs temperature), Isothermal remanent magnetization (IRM), and Hysteresis loop studies were performed using AVFTB (Advanced variable field transition balance) instrument.

4. Results

4.1. Petrography

Polished thin sections were made from one sample in each dyke and examined under an optical microscope and Scanning Electron Microscope (SEM, Hitachi, S-3400N). Representative SEM images are shown in Fig. 2. Most of these samples show ophitic to sub-ophitic texture with

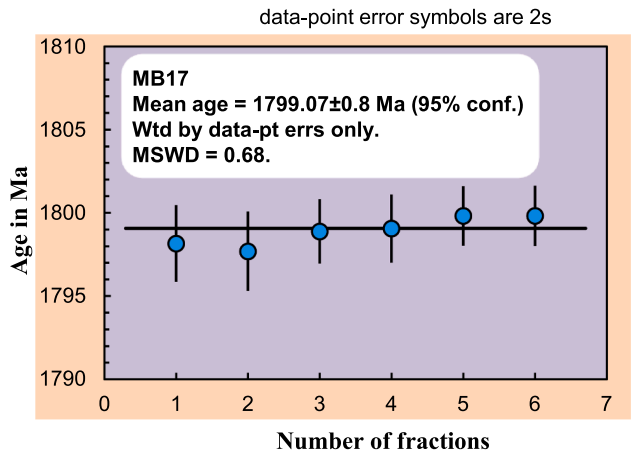


Fig. 3. TE-TIMS weighted mean $^{207}Pb/^{206}Pb$ ages on six baddeleyite fractions from the MB-17 sample (location given in Figure-1). Error bars represent 95% confidence limits of measurements.

plagioclase feldspar and clinopyroxene are major constituents. Magnetite, titanomagnetite, apatite, chlorite, and biotite are observed as opaques. Minor hydrothermal alteration (Sericitization) is observed in plagioclase minerals in most of these samples. Spot analysis was also performed using Energy-Dispersive X-ray Spectroscopy (EDS) to determine the chemical composition of the different minerals.

4.2. Pb-Pb baddeleyite age:

Results of TE-TIMS Pb-Pb baddeleyite isotopic data from the NW-SE trending dyke sample (MB17) are provided in Table 1 and Fig. 3. Six baddeleyite fractions were obtained from this sample. All fractions show similar isotopic ratios (within errors) and give a weighted mean age of 1799.1 ± 0.8 Ma (MSWD = 0.68). This mafic dyke was earlier dated in a site located 8 km northwest of our site MB 17 (present study) near Pebbair village and obtained a U-Pb baddeleyite age of 1794 ± 7 Ma (Söderlund et al., 2019). Combining the two age determinations resulted in a weighted mean age of 1799.0 ± 0.8 Ma (MSWD = 2.1), which probably is the emplacement age of the Pebbair dyke swarm.

4.3. Magnetic mineralogy

Magnetic susceptibility-temperature measurements have been used to determine the magnetic mineralogy in representative samples of the Pebbair dyke swarm. The curves given in Fig. 4, indicate minor thermal alteration in magnetic minerals during heating. The samples show a comparatively large drop in susceptibility at around 575 °C, which suggests that the carrier of magnetic remanence in these dykes is magnetite. The characteristic Isothermal remanent magnetization (IRM) curves for representative samples show magnetic saturation values of about 300 mT and coercive force values lower than 25 mT (Fig. 5). The hysteresis loop studies on the whole rock samples yielded significantly lower magnetization ratios (Mrs/Ms) and low coercivity ratios (Hcr/Hc), which indicates that the dominant magnetic carrier in this dyke swarm is pseudo single domain (PSD) magnetite/ Ti-Magnetite (Fig. 6).

4.4. Paleomagnetism:

Paleomagnetic results from the six dykes are given in Table 2. The mean characteristic remanent magnetization (ChRM) directions of each site are shown in Fig. 7. Examples of typical demagnetization behavior of samples are shown in Fig. 8. The intensity of natural remanent magnetization (NRM) varies from 3 * 10^{-3} A/m to 54 * 10^{-3} A/m. A minimum of five samples from each site were subjected to detailed stepwise AF and/or thermal demagnetization, up to a maximum of 150

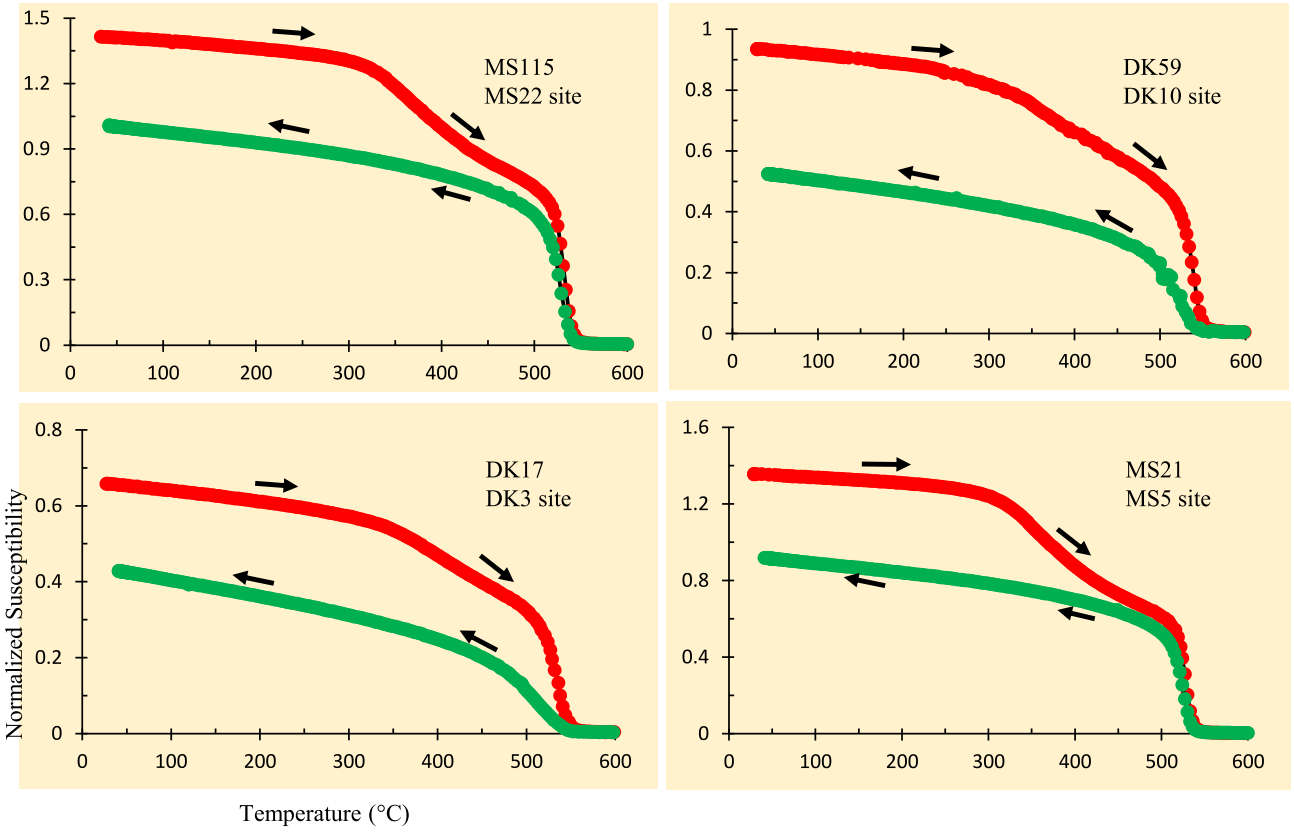


Fig. 4. Typical examples of magnetic susceptibility versus temperature dependencies curves for representative dyke samples from Pebbair dyke swarm.

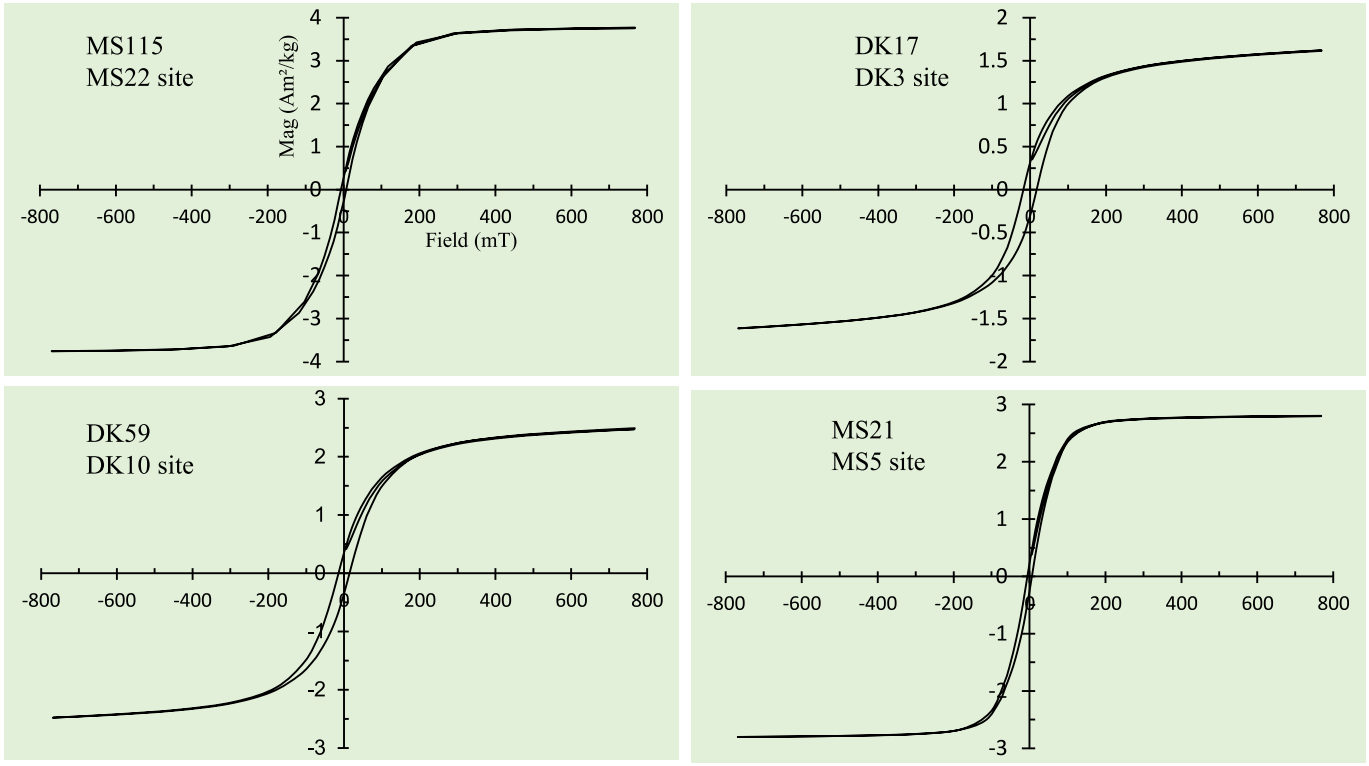


Fig. 5. Typical examples of Hysteresis loop studies on Pebbair dykes show narrow waist-type curves indicating the presence of titanomagnetite or magnetite.

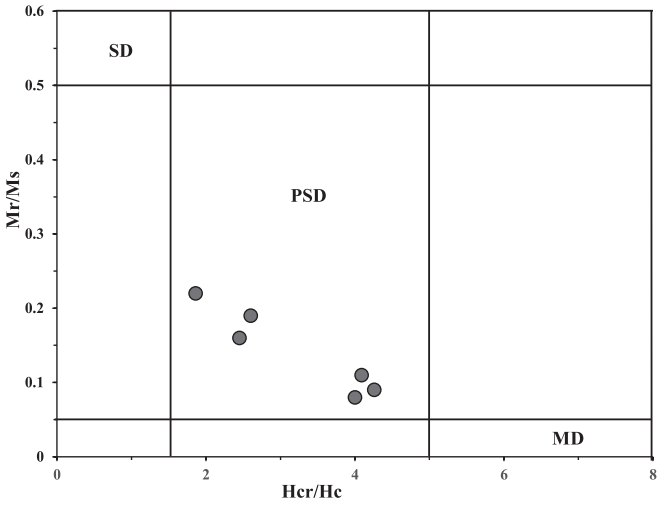


Fig. 6. Magnetic domain characteristics on Day Plot (Day et al., 1977) showing pseudo-single domain magnetite nature for Pebbair dyke swarm.

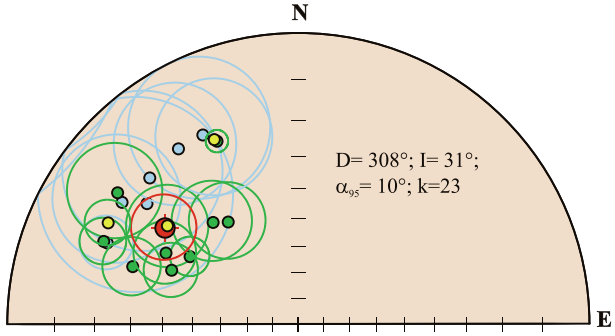


Fig. 7. Stereoplot showing site mean characteristic remanence magnetic directions with ovals of 95% confidence from the ~1.8 Ga dykes. ChRM directions on dated dyke represent by a circle filled with yellow color; the site considered for mean calculation is shown by green-filled circles; excluded sites for mean calculation are shown by blue-filled circles and the mean direction is shown by red-filled circles.

mT or 600 °C. Stable components are carried by high coercivity and blocking temperatures of magnetite and isolated in AF > 40 mT and in the temperature range 540–580° C. The ChRM directions of each site (regardless of their location and orientation) are well clustered, and the mean directions of all sites are overlapped. The directions are displaying northwesterly declination and intermediate positive (downwards) inclination. The mean of the six sites representing six dykes is D = 302°, I = 37°, α_{95} = 10° (k = 47).

Several mafic dykes of variable orientations with similar magnetic remanent directions (within errors) have been reported in previous studies (Piispa et al., 2011; Radhakrishna et al., 2013; Belica et al., 2014; Pivarunas et al., 2018; Parashuramulu et al., 2021; Miller et al., 2023). The site locations are given in Table 2 and shown in Fig. 1. Most of these dykes also occur in the vicinity of the Cuddapah basin with dominant trends of E-W and N-W. However, following our acceptance criteria (α_{95} < 15°), five sites (1, 30, 42, DK10, and 1a) were accepted for grand mean calculation, and six sites (P10, P64, P25, P71, NM, and P9) were rejected. One site (PV10) of the mafic sill, intruded into the Cuddapah basin also show a similar magnetization direction, which is a part of sill intrusion, dated at ~1.89 Ga (French et al., 2008; Sarma et al., 2020). Here, we suspect this could be an overprinted direction, and not considered for further calculation. An overall mean remanence direction calculated for 11 sites, including the present study together with reported sites yields D = 306°, I = 31°, α_{95} = 10° (k = 23). The corresponding paleomagnetic pole is calculated at 38°N, 357°E (dp = 6.1° and dm = 10.9°) (Fig. 9, Table 2).

The new paleomagnetic pole reporting here fulfills five reliability criteria out of seven of Van der Voo (1990) and Meert et al. (2020) as

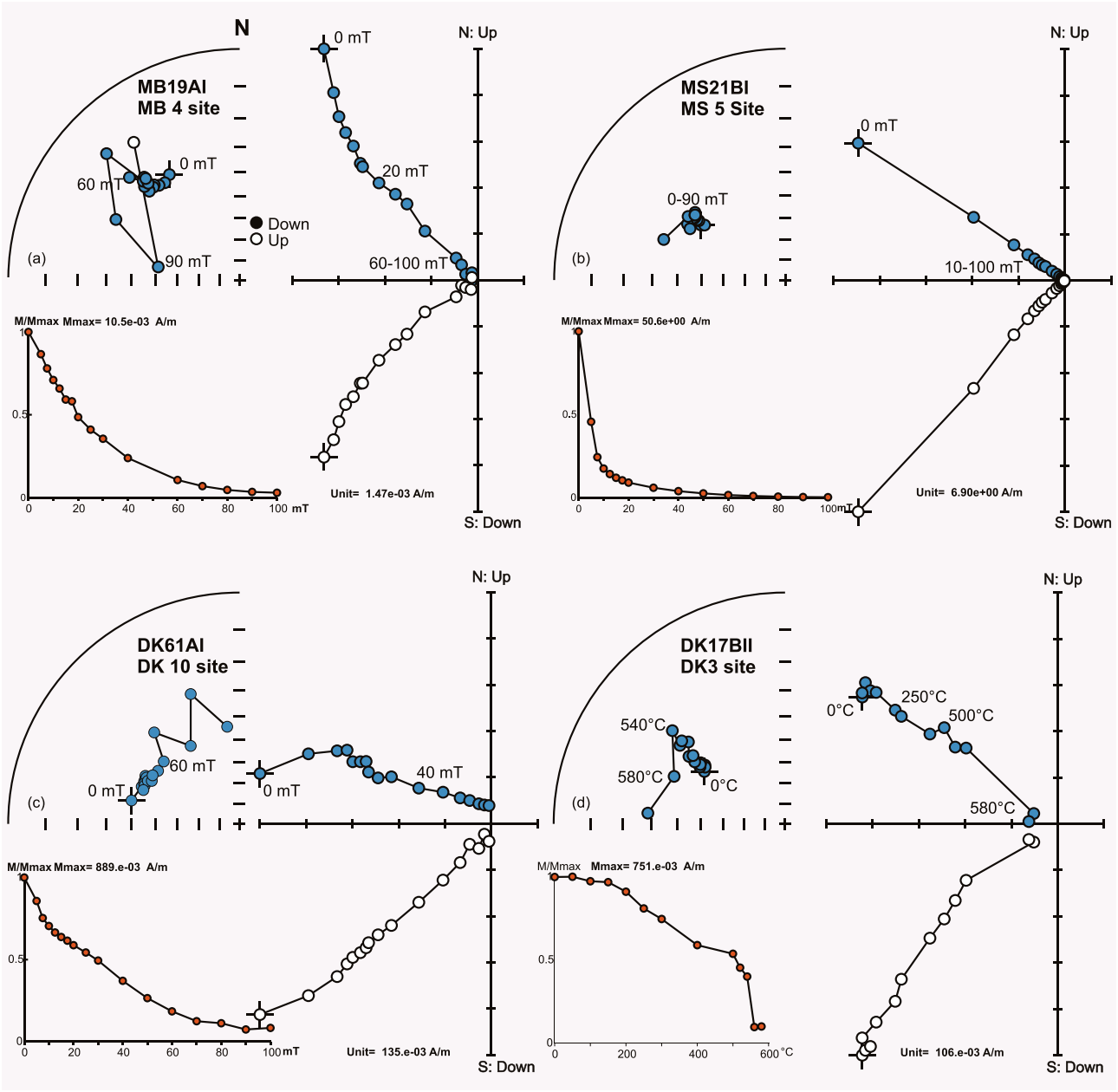


Fig. 8. Zijderveld diagrams and equal area stereonet projections showing characteristic behavior of natural remanence to AF demagnetization (a to c) and thermal demagnetization (d) for representative samples of Pebbair dyke swarm. AF measurements are in millitesla (mT) and Thermal demagnetization measurements are in °C. Open/closed circles in the stereo plots represent upward/downward directed vectors and open/closed circles in the Zijderveld plots represent vertical/horizontal projections. Plotted using Remasoft 3.0, a plotting and analysis program (Chadima and Hrouda, 2006).

described below.

(1) The mean paleomagnetic pole yields well-determined ages of 1799.1 ± 0.8 Ma (TE-TIMS Pb-Pb baddeleyite, present study) and 1794 ± 7 Ma (ID-TIMS, U-Pb baddeleyite, Söderlund et al., 2019). This is inferred as the age of magnetization.
(2) A total of 104 samples from 11 sites revealed the new pole, and show the Fisher statistical parameters k = 23 and α_{95} = 9.7°.
(3) The ChRM directions were separated with least squares fit from zijderveld diagrams (Zijderveld, 1967), and statistical calculations were done by principal component analysis (Kirschvink, 1980) after standard alternating field (AF) and thermal demagnetization (TH) procedures.
(4) We were unable to obtain a positive field test during this study. However, the paleosecular variation (PSV) test on eleven sites (N = 11) shows the 95% confidence circle of the mean paleomagnetic pole (A95 = 8.1°) falls within the reliability envelope (A_{95} min = 4.6°, A_{95} max = 18.1°) (Deenen et al., 2011). This indicates these results have well-sampled and adequate averaging of PSV for this new paleomagnetic pole.
(5) The mafic dykes of the present study are not deformed or metamorphosed, and no significant titling and rotation have been reported for this period (Pivarunas and Meert, 2020).
(6) No magnetic reversal has been recorded in this dyke swarm.
(7) The obtained pole has no resemblance to known younger-age paleomagnetic poles for this craton.

5. Discussion

The paleogeographic position of India within Paleoproterozoic

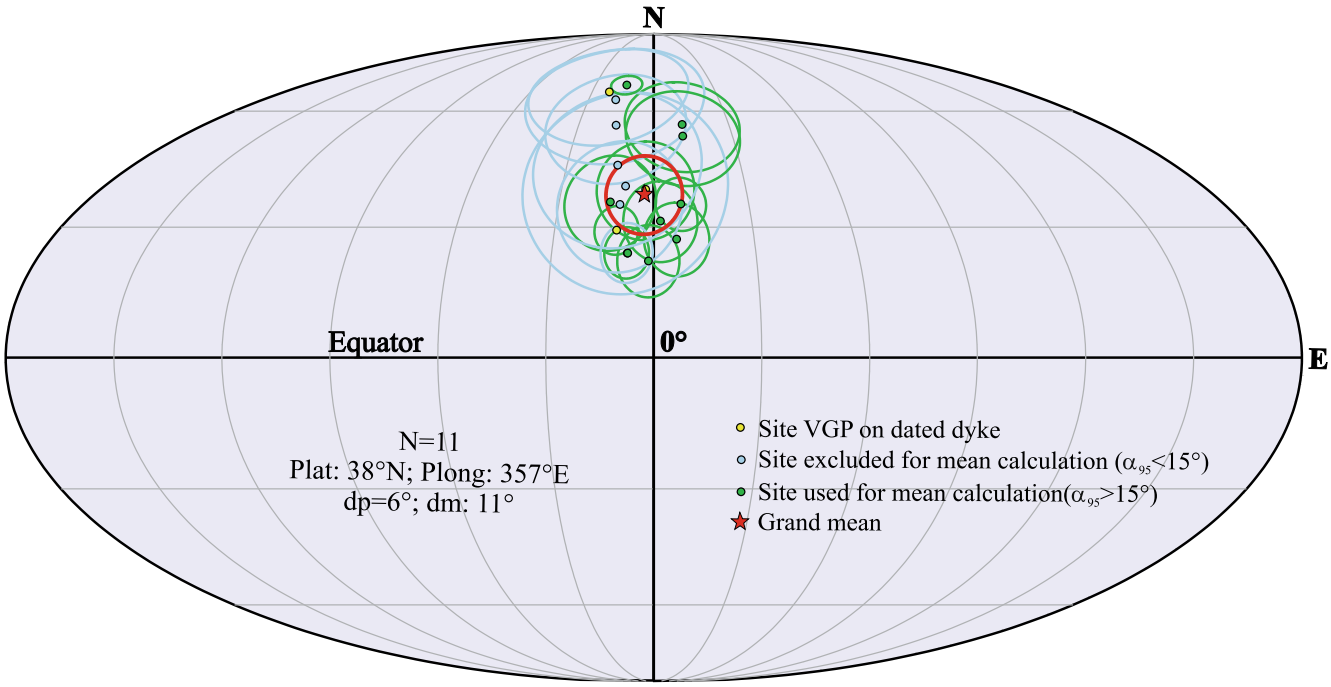


Fig. 9. Mollweide projection showing the virtual geomagnetic poles of the site means from the ~1.8 Ga Pebbair dyke swarm.

supercontinents configurations has been examined using highly reliable key poles from ~ 2.37 Ga to ~1.86 Ga in recent studies (Meert et al., 2010; Radhakrishna et al., 2013; Belica et al., 2014; Nagaraju et al., 2018b; Parashuramulu et al., 2021; Meert and Santosh, 2022). In this study, we report a new ~1.8 Ga key pole at 38°N, 357°E (A_{95} = 8°) for the Pebbair dyke swarm. Coeval mafic units are not known from any region of the DC and other cratons of India. WNW-ESE trending dykes of Pipilia swarm in Singhbhum craton have been dated at 1766 ± 1 Ma and 1764.6 ± 0.9 Ma (Shankar et al., 2014) is the only marginally correlatable younger mafic event in peninsular India. Srivastava et al. (2018), suggested a possible relation between Dharwar and Singhbhum craton as proximal to each other at ~1.77–1.79 Ga in the part of Pipilia-Pebbair LIP. In addition, Rb-Sr age of 1815 ± 200 Ma has been reported for Gwalior upper series lavas (Crawford and Compston, 1969). Recently, in-situ SIMS U-Pb baddeleyite age of 1733 ± 23 Ma has been reported for the mafic sill intrusions within the Gwalior basin (Srivastava et al., 2022), and paleomagnetic directions were reported on Gwalior sills by Pradhan et al, 2010. The calculated drift rate of DC from ~1.86 Ga to ~1.80 Ga suggests that the DC drifted northward with a minimum velocity of ~5.7 cm/year during that period.

5.1. Global mafic magmatism at ~1.8 Ga

Contemporaneous magmatic episodes are known from a few cratons around the globe. Potentially correlative units include two gabbro units in Speewah sill (ID-TIMS baddeleyite, 1791.7 ± 4.1 Ma and 1792.7 ± 1.3 Ma; Ramsay et al., 2019) and mafic pegmatite vein in the Hart Dolerite (SHRIMP baddeleyite, 1792.1 ± 5.9 Ma; Ramsay et al., 2019); granophyre sample of Hart dolerite (1799 ± 17 Ma, 1795 ± 15 Ma; Sheppard et al., 2012) from the north Australia craton. U-Pb baddeleyite ages of Avanavero dolerite (1794.5 ± 1.6 Ma, Reis et al., 2013), Quarenta llhas dolerite (1792.6 ± 1.3 Ma, Reis et al., 2013), Omai gabbro dyke (1794 ± 4 Ma, Norcross et al., 2000); Manga brava sill dolerite (1782 ± 3 Ma, Santos et al., 2003); Cipo sill (1787 ± 14 Ma, Santos et al., 2003) of the Guiana shield, Amazonian craton. NW trending dykes of the Pará de Minas swarm yield U-Pb baddeleyite ages of 1798 ± 4 Ma, 1791 ± 7 Ma, and 1793 ± 18 Ma (Cederberg et al., 2016) from the São Francisco craton, South America. The ENE striking tholeiitic Florida dyke swarm (U-Pb baddeleyite, 1790 ± 5 Ma, Halls et al., 2001) of the Rio de la Plata craton. U-Pb baddeleyite age of the Tomashgorod dolerite dyke (1790.7 ± 5.1 Ma and 1787.4 ± 6.4 Ma; Bogdanova et al. 2013; Shumlyanskyy et al., 2012 respectively); Gabbro from the Davydky

Table 3
Paleomagnetic poles used for paleogeographic reconstruction at ~1.8 Ga.

Craton/Continent	Mafic event	Age (Ma)	Reference	P_{lat} (°N)	P_{long} (°E)	A_{95} (°)	Reference	Euler pole (λ; Φ; ω)	Rlat (°N)	Rlong (°E)
Dharwar craton/ India	Pebbair dyke swarm	1799–1794	Present study; Söderlund et al. (2019)	38.0	357	8	Present study	13.72; −86.16; 53.63	89.98	1.03
North Australia	Hart Dolerite sills	1793–1792	Ramsay et al. (2019)	−5.3	181.0	12	Kirsher et al., 2019	10.35; 108.12; 100.06	84.39	93.32
Rio de la Plata craton	Florida mafic dykes	1790 ± 5	Teixeira et al. (2013)	−77.5	162.0	10.2	Teixeira et al. (2013)	−5.95; 39.31; 164.52	82.88	83.82
São Francisco Craton	Pará de Minas dykes	1798–1791	Cederberg et al. (2016)	−39.8	196.8	17	D'Agrella-Filho et al. (2020)	–22.29; 43.77; −189.163	77.91	−52.09
Amazonian craton	Avanavero mafic sills	1793–1795	Reis et al. (2013)	48.4	207.9	9.6	Bispo-Santos et al. (2014)	68.97; −170.45; 144.45	88.31	–23.19
Baltica-Fennoscandia	Mean Hoting gabbro	1786 ± 10	Hellström et al. (2003)	43	233.3	10.9	Elming et al. (2009)	55.24; 162.49; 48.64	70.14	−84.61

P_{lat}: pole latitude; Pl_{ong}: pole longitude; A_{95}: confidence circle - Fisher's statistic parameter; λ and Φ are the latitude and longitude of Euler poles respectively; ω is the Euler angle used for rotation; ω is positive for counter-clockwise rotation and negative for clockwise rotation; Rlat: rotated latitude; Rlong: rotated longitude.

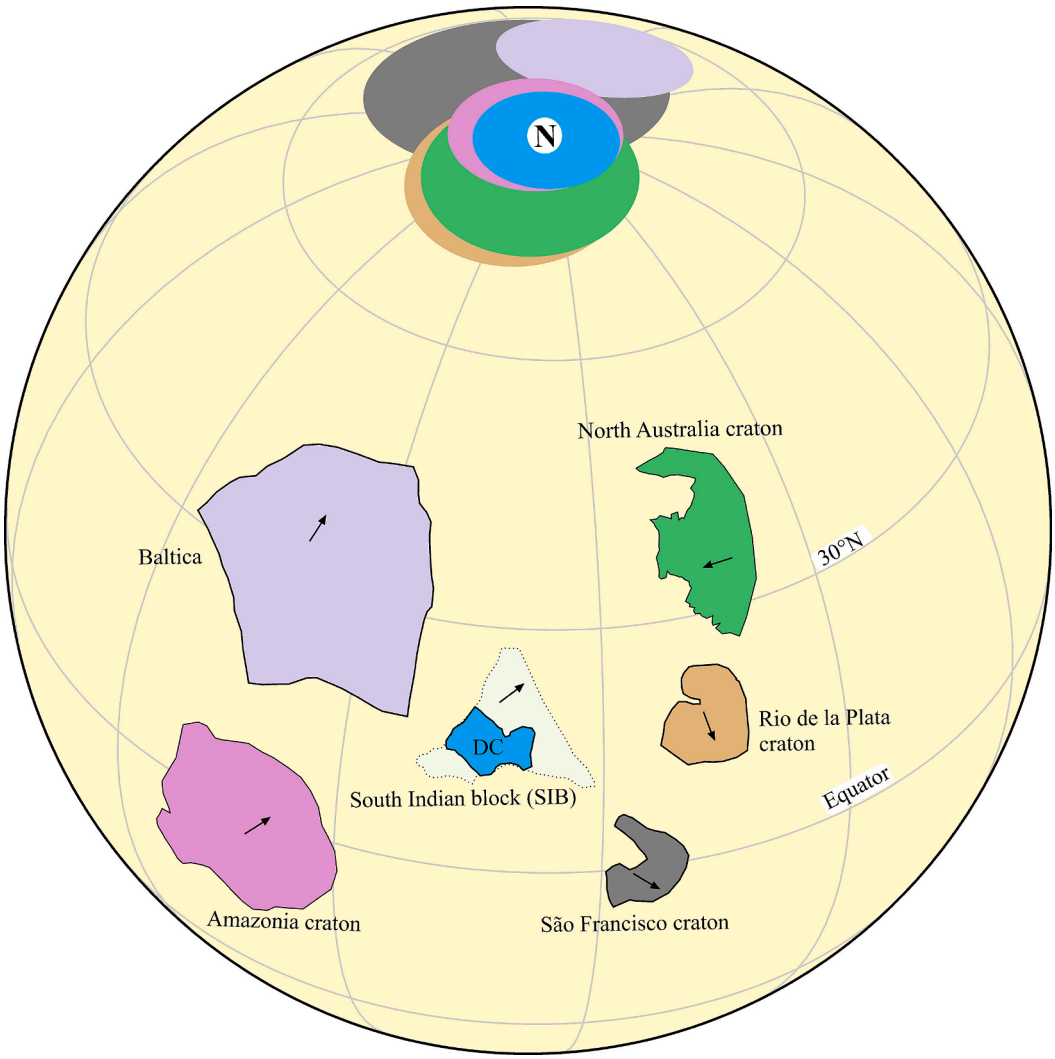


Fig. 10. Paleogeographic reconstruction at ~1.8 Ga has been attempted utilizing key paleopoles from different cratons/continents, given in Table- 3. Respective Euler rotation parameters (to the absolute framework) used in these reconstructions are also given in Table 3. The pole and craton/continent pairs are depicted by the same color combination. The black arrow demonstrates the present-day North for individual cratons/continents.

massif (1789.8 ± 1.5 Ma, Shumlyanskyy et al., 2017); Rudnya Bazarska dyke (1793 ± 3 Ma; Shumlyanskyy et al., 2016) and Zamyslovychi dyke (1789 ± 9 Ma; Shumlyanskyy et al., 2016) in the Ukrainian shield. NW-SE trending Ussiranniak Lake dyke (1799.6 ± 4.1 Ma; Sahin and Hamilton 2019) from the North Atlantic craton.

5.2. Paleogeographic reconstruction at ~1.8 Ga

Well-defined paleomagnetic poles along with robust geochronology (U-Pb or Pb/Pb Baddeleyite or Zircon) of coeval mafic intrusives are considered for paleogeographic reconstruction model at ~1.8 Ga. The paleopoles used from other cratons are summarized in Table 3 and the paleogeographic reconstruction model is given in Fig. 10. The presented reconstruction here is based on a perceived geometric fitting of coeval mafic magmatic intrusions such as dykes and sills. Here, we use the 'closest approach' technique (Meert and Torsvik, 2003) to bring the cratons into position in our reconstruction. The paleolatitude and azimuthal orientation of each craton/continent is fixed but the paleo-longitude and polarity are ambiguous (Meert and Torsvik, 2003; Buchan, 2014). The Euler rotation parameters for the respective paleopoles of each craton are given in Table 3. The paleogeographic reconstruction model is given in Fig. 10 and was prepared using the GPLATES software (https://www.gplates.org/).

The assembly of the North (Aravalli-Delhi-Bundelkhand) and South (Dharwar-Bastar-Singhbhum) India blocks was taken place during ~1.06–0.93 Ga to form a final configuration of the Greater Indian landmass (Bhowmik et al., 2012; Meert and Pandit, 2015; Bhowmik, 2019; Meert et al., 2021), so we have considered the South India block outline for reconstruction. The North Australian craton was assembled in the later stages of the amalgamation of crustal blocks (Kimberly craton) at about ~1800 Ma (Ramsay et al., 2017). The São Francisco/Congo craton was formed by the collision belts connecting Archean blocks at about ~2.0 Ga (Teixeira et al., 2019). The Rio de la Plata craton is evolved by an accretionary orogeny at about ~2.3–2.0 Ga (Teixeira et al., 2013). The Baltica craton was assembled in its present-day configuration at 1900–1700 Ma (Bogdanova et al., 2008; Elming et al., 2010).

In Fig. 10, the paleoposition of the Dharwar craton (South India block) is placed at shallow paleolatitude (above the equator) by the paleopole of the Pebbair dyke swarm (present study) at ~1.8 Ga. The North Australia craton was located at moderate paleolatitude with the paleopole of Hart dolerite sills (Kirscher et al., 2019). The paleoposition of the Rio de la Plata craton is at shallow latitude based on the paleopoles of Florida mafic dykes (Teixeira et al., 2013). The São Francisco craton and the Amazonia craton are occupied around the equator Pará de Minas dykes (D'Agrella-Filho et al., 2020) and Avanavero mafic sills

(Bispo-Santos et al., 2014) respectively. The Baltica craton was located at moderate paleolatitude with the paleopole of Hoting gabbro (Elming et al., 2009).

The Columbia supercontinent assembled at ~1.75 Ga (Zhang et al., 2012; Meert and Santosh, 2017; Meert et al., 2021; Meert and Santosh, 2022). In the various paleogeographic models for the Columbia supercontinent, the exact position/orientation of India is still a debatable topic. As per Zhao et al. (2002), India lies between the North China craton and Australia. Whereas, India has been proposed to be in proximity to West Africa (which was attached to Amazonia at ~2 Ga, D'Agrella-Filho et al., 2021), North China craton, and Siberia (Zhang et al., 2012). Another model proposed India be positioned in the vicinity of the Baltica, São Francisco, and Kalahari craton (Pisarevsky et al., 2014). A recent model proposed India's paleoposition between proto-Australia and Siberia (Chaves and de Rezende, 2019).

Based on the above discussion, the cratons/continents which were potentially proximal to the South India block in the Columbia supercontinent were North China, Australia, Amazonia, Serbia, Baltica, São Francisco, and Kalahari. So, it is very likely that in the pre-Columbia era i.e., before ~1.75 Ga, the South India block might have been positioned next to some of these cratons/continents. The ~1.80 Ga paleogeographic reconstruction proposed here is a pre-Columbia assembly as it predates the Columbia supercontinent assembly by ~50 Ma.

In the reconstruction presented here (Fig. 10), the Dharwar (South India block), Amazonia, São Francisco, Rio de la Plata, Baltica, and North Australia cratons are proposed to lie nearby between the equator and 45°N latitude. The longitudes are not fixed here. Another pre-Columbia paleogeographic reconstruction at ~1.86 Ga (Fig. 11 of Parashuramulu et al., 2021) placed the Dharwar (South India block), Amazonia, Baltica, and Siberia cratons nearby at the equator and 30°S latitude. So, based on these, we propose that the South India block, Amazonia, and Baltica have assembled at ~1.86 Ga and at ~1.80 Ga the North Australia, São Francisco, and Rio de la Plata cratons came in close to the South India block, Amazonia, and Baltica assembly. The presence of coeval mafic dykes/sills emplaced in these cratons at ~1.86 Ga (Parashuramulu et al., 2021 references therein) and ~1.80 Ga (this study) extends the geological evidence in support of the proposed assembly. To confirm whether these mafic dykes/sills have the same source, further studies such as geochemical and isotopic studies will be required.

6. Conclusion

1. We present TE-TIMS Pb-Pb baddeleyite age of 1799.1 ± 0.8 Ma for NW-SE trending dolerite dyke in the northwestern region of the Cuddapah basin. This new age is matching (within errors) with an earlier reported U-Pb baddeleyite age of 1794 ± 7 Ma (Söderlund et al., 2019), and gives a weighted mean age of 1799.0 ± 0.8 Ma (MSWD = 2.1) for the emplacement of Pebbair dyke swarm.
2. Paleomagnetic results from six sites of the present study and five sites from previous studies reveal a mean remanent direction D = 306°, I = 31°, α_{95} = 10° (k = 23), and corresponding paleopole at 38°N, 357°E (dp = 6.1° and dm = 10.9°).
3. This new pole satisfies five reliability criteria (R = 5) of Van der Voo (1990) and Meert et al., (2020), and passes the paleosecular variation (PSV) test, and is considered a key paleomagnetic pole at ~1.8 Ga for Dharwar craton.
4. The Dharwar craton drifted northward from ~1.86 Ga to ~1.80 Ga with a minimum velocity of ~5.7 cm/year.
5. The Dharwar craton (South India block), the Amazonia craton, and the Baltica craton have assembled at ~1.86 Ga, and North Australia, São Francisco and Rio de la Plata cratons came in close to the South India block, Amazonia, and Baltica assembly at ~1.80 Ga..

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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